

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) — Semester V Examination, 2016-17

Paper No. GLB-501: Physical and Structural Geology

Time: 3 Hours

Maximum Marks: 70

Ref. No.: BS/Sem V/175

*Instructions: Attempt any **five** questions in all, at least **two** questions each from Section A and B. Section C is compulsory. All questions carry equal marks (14 marks each).*

SECTION — A

Question 1. Describe the classification of folds based on dip-isogons. [14]

Question 2. Describe the erosional and depositional features formed by rivers. [14]

Question 3. Discuss with suitable examples: [2 × 7 = 14]

- (a) Types of sand dunes
- (b) Mitigation of landslides

Question 4. What is an unconformity? Describe the different types of unconformity and explain how you would recognize an unconformity in the field. [14]

SECTION — B

Question 5. Write detailed notes on the following: [2 × 7 = 14]

- (a) Effect of faulting on folded outcrop
- (b) Planar and linear structures in deformed rocks

Question 6. Write short notes on the following: [4 × 3.5 = 14]

- (a) Pratt's concept of Isostasy
- (b) Erosional geomorphic features of the ocean
- (c) Divergent plate boundary
- (d) Plateau

Question 7. Discuss the following: [2 × 7 = 14]

- (a) Endogenic processes
- (b) Joints and their geometrical classification

Question 8. Differentiate between the following:

[4 × 3.5 = 14]

- (a) Window and Klippe
- (b) Horst and Graben
- (c) Tension joint and shear joint
- (d) Fault and unconformity

SECTION — C (Compulsory)

Question 9. Describe each of the following in two to three sentences. Give suitable diagrams where applicable:

[7 × 2 = 14]

- (i) Geomagnetic time scale
- (ii) Difference between till and moraine
- (iii) Angle of repose
- (iv) Allochthonous tectonic unit
- (v) Flower structure
- (vi) Similar fold and Parallel fold
- (vii) Antiformal Syncline